**A**

**Blood expression instruments**
4,462 potentially druggable genes
2,544 genes with instruments
6,135 SNP–gene records

→

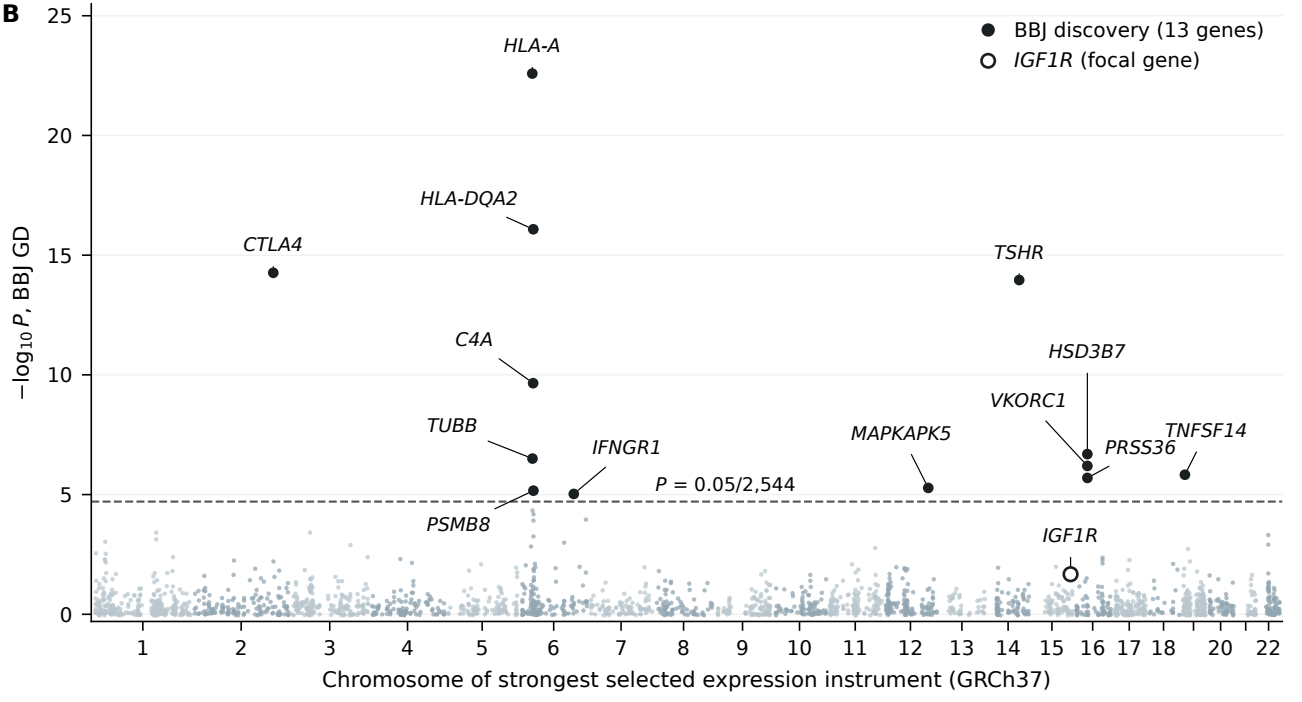
**BBJ GD discovery**
2,234 estimable genes
13 discoveries
Threshold $P < 0.05/2{,}544$

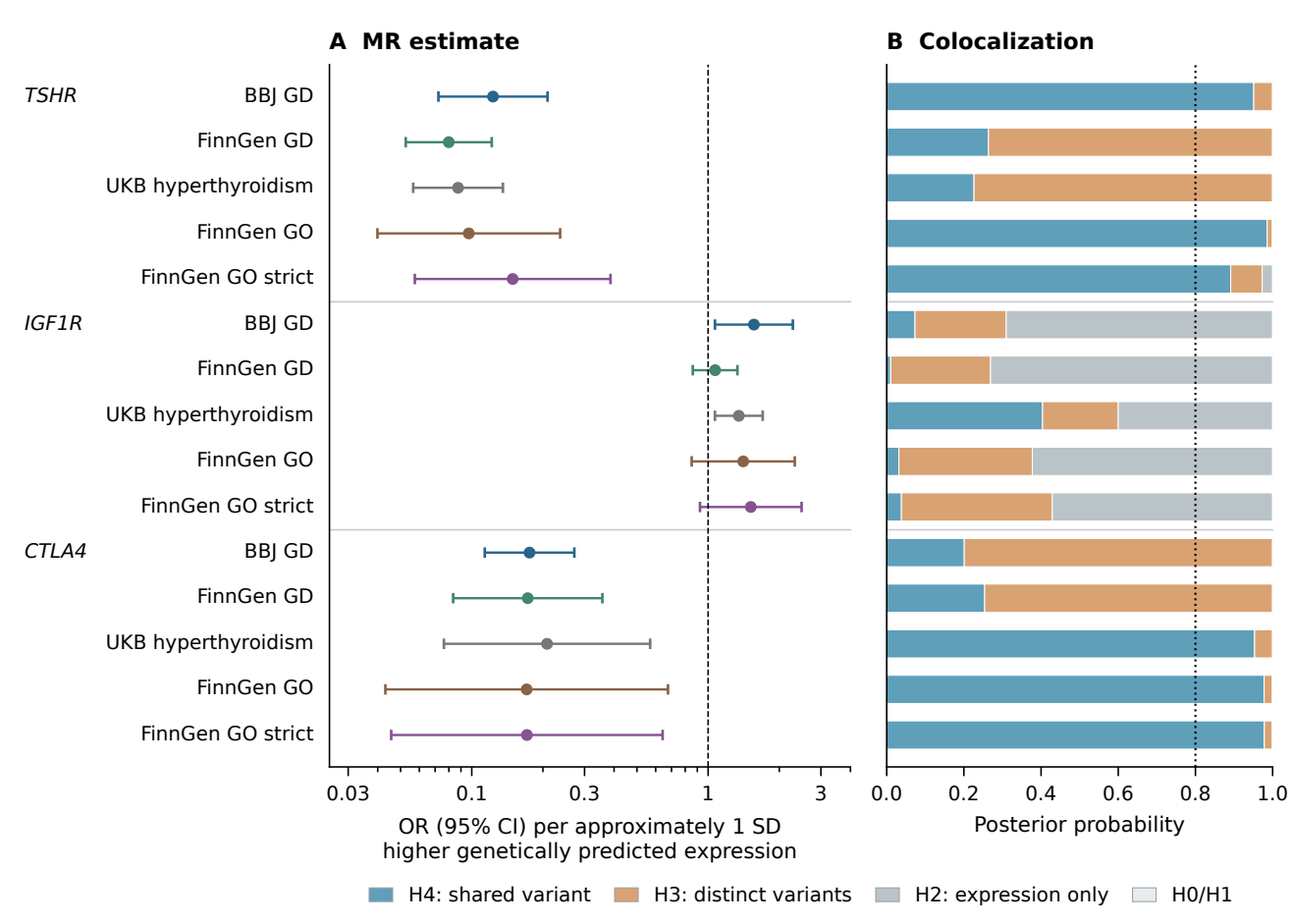
→

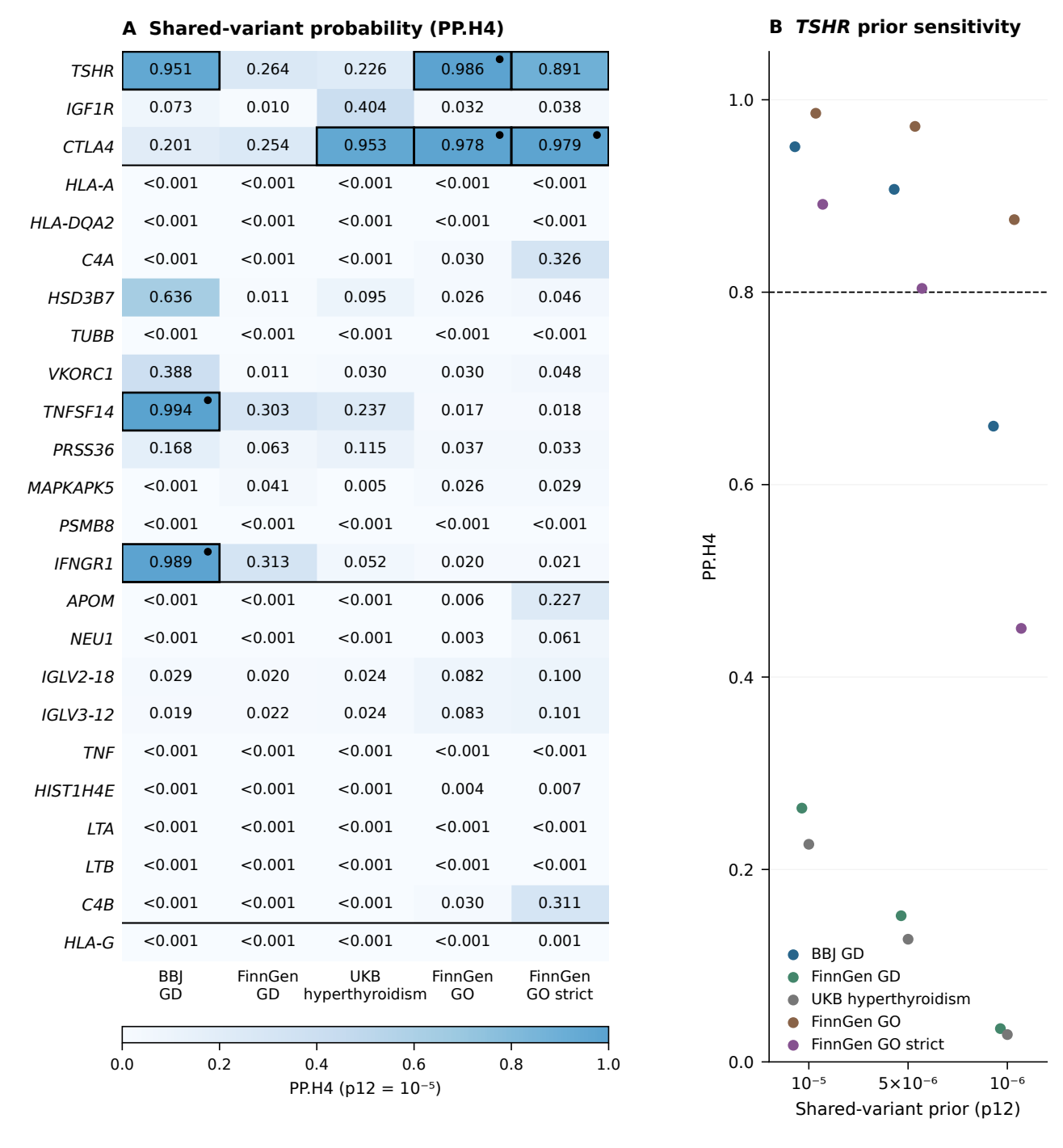
**Comparison datasets**
UKB hyperthyroidism, FinnGen GO
Post hoc: FinnGen GD,
FinnGen GO strict, GTEx tissues

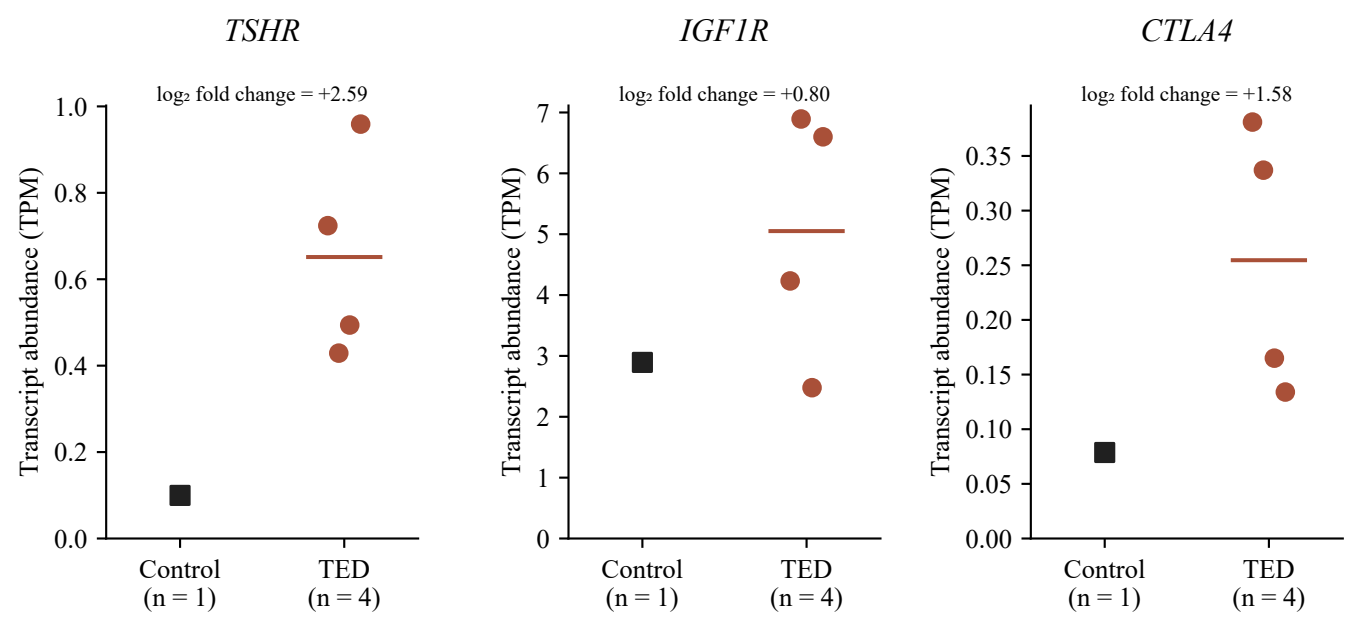
Colocalization: 24 selected genes, centered on *TSHR* and *IGF1R*, with *CTLA4* as comparator

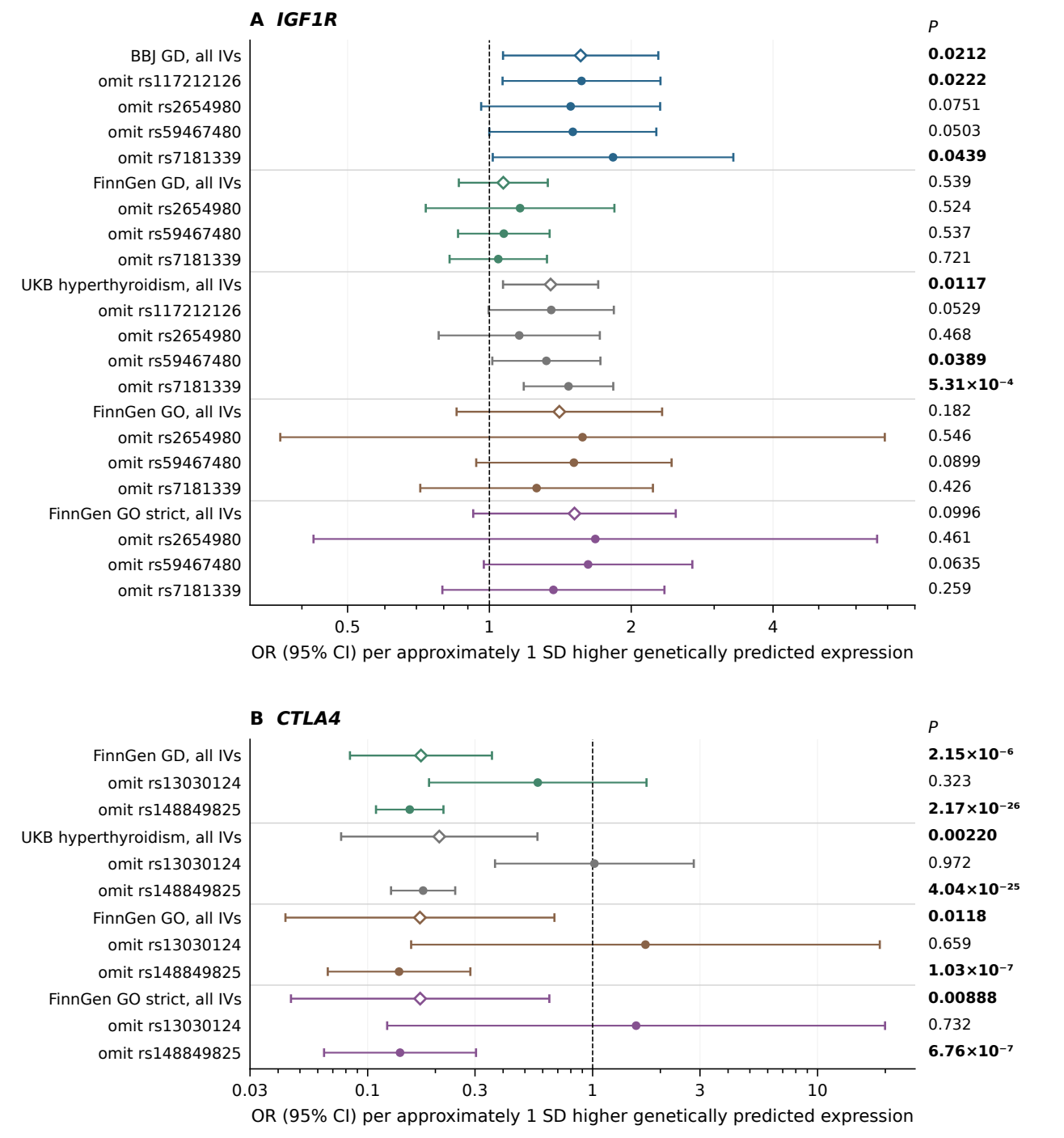
**B**




A MR estimate
B Colocalization
TSHR
IGF1R
CTLA4
BBJ GD
FinnGen GD
UKB hyperthyroidism
FinnGen GO
FinnGen GO strict
0.03
0.1
0.3
1
3
OR (95% CI) per approximately 1 SD higher genetically predicted expression
0.0
0.2
0.4
0.6
0.8
1.0
Posterior probability
H4: shared variant
H3: distinct variants
H2: expression only
H0/H1


A Shared-variant probability (PP.H4)
TSHR 0.951 0.264 0.226 0.986 0.891
IGF1R 0.073 0.010 0.404 0.032 0.038
CTLA4 0.201 0.254 0.953 0.978 0.979
HLA-A <0.001 <0.001 <0.001 <0.001 <0.001
HLA-DQA2 <0.001 <0.001 <0.001 <0.001 <0.001
C4A <0.001 <0.001 <0.001 0.030 0.326
HSD3B7 0.636 0.011 0.095 0.026 0.046
TUBB <0.001 <0.001 <0.001 <0.001 <0.001
VKORC1 0.388 0.011 0.030 0.030 0.048
TNFSF14 0.994 0.303 0.237 0.017 0.018
PRSS36 0.168 0.063 0.115 0.037 0.033
MAPKAPK5 <0.001 0.041 0.005 0.026 0.029
PSMB8 <0.001 <0.001 <0.001 <0.001 <0.001
IFNGR1 0.989 0.313 0.052 0.020 0.021
APOM <0.001 <0.001 <0.001 0.006 0.227
NEU1 <0.001 <0.001 <0.001 0.003 0.061
IGLV2-18 0.029 0.020 0.024 0.082 0.100
IGLV3-12 0.019 0.022 0.024 0.083 0.101
TNF <0.001 <0.001 <0.001 <0.001 <0.001
HIST1H4E <0.001 <0.001 <0.001 0.004 0.007
LTA <0.001 <0.001 <0.001 <0.001 <0.001
LTB <0.001 <0.001 <0.001 <0.001 <0.001
C4B <0.001 <0.001 <0.001 0.030 0.311
HLA-G <0.001 <0.001 <0.001 <0.001 0.001
BBJ GD
FinnGen GD
UKB hyperthyroidism
FinnGen GO
FinnGen GO strict
0.0 0.2 0.4 0.6 0.8 1.0
PP.H4 (p12 = 10⁻⁵)
B TSHR prior sensitivity
PP.H4
1.0 0.8 0.6 0.4 0.2 0.0
BBJ GD
FinnGen GD
UKB hyperthyroidism
FinnGen GO
FinnGen GO strict
10⁻⁵ 5×10⁻⁶ 10⁻⁶
Shared-variant prior (p12)


TSHR
log₂ fold change = +2.59
Transcript abundance (TPM)
1.0
0.8
0.6
0.4
0.2
0.0
Control
(n = 1)
TED
(n = 4)
IGF1R
log₂ fold change = +0.80
Transcript abundance (TPM)
7
6
5
4
3
2
1
0
Control
(n = 1)
TED
(n = 4)
CTLA4
log₂ fold change = +1.58
Transcript abundance (TPM)
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
Control
(n = 1)
TED
(n = 4)


A IGF1R
P
BBJ GD, all IVs
omit rs117212126
omit rs2654980
omit rs59467480
omit rs7181339
FinnGen GD, all IVs
omit rs2654980
omit rs59467480
omit rs7181339
UKB hyperthyroidism, all IVs
omit rs117212126
omit rs2654980
omit rs59467480
omit rs7181339
FinnGen GO, all IVs
omit rs2654980
omit rs59467480
omit rs7181339
FinnGen GO strict, all IVs
omit rs2654980
omit rs59467480
omit rs7181339
0.0212
0.0222
0.0751
0.0503
0.0439
0.539
0.524
0.537
0.721
0.0117
0.0529
0.468
0.0389
5.31×10⁻⁴
0.182
0.546
0.0899
0.426
0.0996
0.461
0.0635
0.259
0.5
1
2
4
OR (95% CI) per approximately 1 SD higher genetically predicted expression
B CTLA4
P
FinnGen GD, all IVs
omit rs13030124
omit rs148849825
UKB hyperthyroidism, all IVs
omit rs13030124
omit rs148849825
FinnGen GO, all IVs
omit rs13030124
omit rs148849825
FinnGen GO strict, all IVs
omit rs13030124
omit rs148849825
2.15×10⁻⁶
0.323
2.17×10⁻²⁶
0.00220
0.972
4.04×10⁻²⁵
0.0118
0.659
1.03×10⁻⁷
0.00888
0.732
6.76×10⁻⁷
0.03
0.1
0.3
1
3
10
OR (95% CI) per approximately 1 SD higher genetically predicted expression